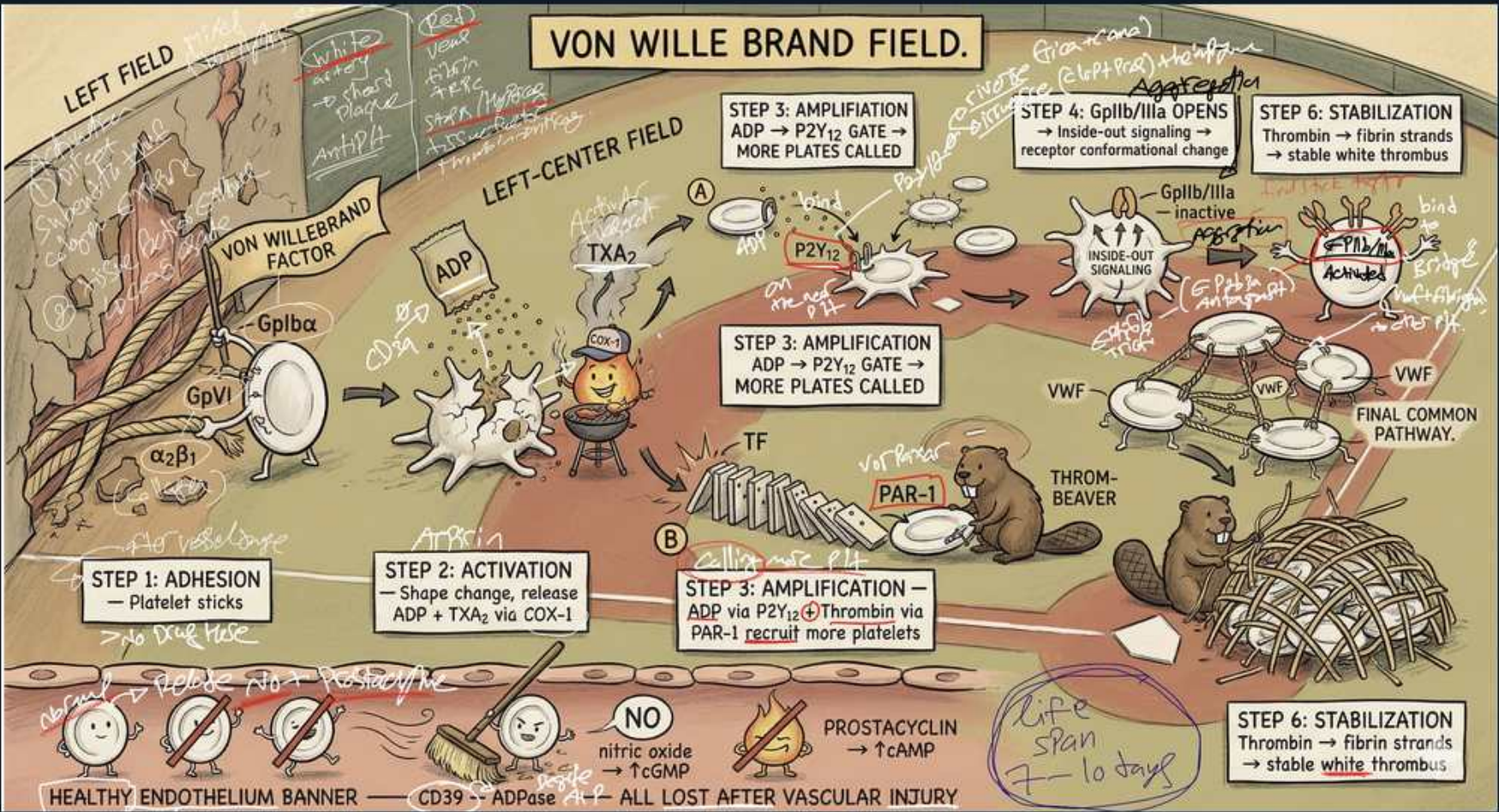


Antiplatelets — Hemostasis Cascade & Drug Targets (Von Wille Brand Field)



1. GPIb-alpha + Von Willebrand Factor (adhesion)

WHAT YOU SEE

A platelet grabbing a thick rope labeled Von Willebrand Factor via a hook labeled GpIbalpha at exposed subendothelium.

WHAT IT ENCODES

STEP 1 ADHESION: after vessel injury, subendothelial collagen exposes VWF, which bridges GPIb-alpha on the resting platelet to the wall under high shear. This is the very first tethering step. Her note 'no drug here' flags that no routine antiplatelet acts at GPIb/VWF adhesion.

BOARD TRAP

Board may claim aspirin or clopidogrel block initial adhesion — wrong. Adhesion is GPIb-VWF (and GPVI/alpha2beta1 to collagen); standard antiplatelets act downstream at TXA2, P2Y12, or GPIIb/IIIa, not at the VWF tether.

2. GPVI & alpha2beta1 (collagen receptors)

WHAT YOU SEE

Platelet feet labeled GpVI and alpha2beta1 planted on broken collagen rubble.

WHAT IT ENCODES

Collagen binds directly via GPVI (signaling) and alpha2beta1 (firm anchorage), reinforcing adhesion and triggering activation. GPVI is the key collagen signaling receptor that helps shape-change begin.

BOARD TRAP

Don't confuse alpha2beta1 (collagen, integrin) with alphaIIbbeta3 = GPIIb/IIIa (fibrinogen, final aggregation). Tirofiban/eptifibatide/abciximab block GPIIb/IIIa, NOT the collagen receptors.

3. ADP granule release

WHAT YOU SEE

A bag bursting and spilling ADP pellets onto the activated spiky platelet.

WHAT IT ENCODES

STEP 2 ACTIVATION: activated platelets degranulate and release ADP from dense granules, which feeds the P2Y12 amplification loop. ADP is the autocrine/paracrine signal that recruits and activates more platelets.

BOARD TRAP

ADP acts on P2Y12 (and P2Y1) — clopidogrel/prasugrel/ticagrelor target P2Y12. A trap is attributing ADP signaling to the TXA2/COX pathway (that's aspirin's lane).

4. TXA2 + COX-1 (the grill)

WHAT YOU SEE

A fiery grill chef labeled COX-1 cooking up smoke labeled TXA2.

WHAT IT ENCODES

STEP 2: activation triggers COX-1 to convert arachidonic acid to thromboxane A2 (TXA2), a potent secreted agonist that amplifies aggregation and causes vasoconstriction. This is exactly aspirin's target.

BOARD TRAP

Aspirin IRREVERSIBLY acetylates COX-1, so the effect lasts the platelet lifespan (~7-10 days) because platelets have no nucleus to resynthesize COX-1. NSAIDs (ibuprofen) reversibly block COX and can competitively block aspirin's access.

5. P2Y12 gate (amplification)

WHAT YOU SEE

Circled label P2Y12 acting as a gate that calls 'more plates' (more platelets).

WHAT IT ENCODES

STEP 3 AMPLIFICATION: ADP acting on the P2Y12 receptor sustains activation, stabilizes the aggregate, and recruits additional platelets. P2Y12 is the central amplification node for thienopyridines and ticagrelor.

BOARD TRAP

Clopidogrel and prasugrel are irreversible prodrugs needing CYP activation; ticagrelor is a reversible, direct (non-prodrug) P2Y12 blocker. Mixing up reversibility/prodrug status is the classic discriminator.

6. PAR-1 / thrombin (THROM-BEAVER)

WHAT YOU SEE

A beaver labeled THROM-BEAVER beside PAR-1, near the tissue-factor woodpile (TF).

WHAT IT ENCODES

STEP 3: thrombin (from the TF/coagulation cascade) is the most potent platelet agonist, acting through PAR-1 to amplify aggregation and recruit platelets. Vorapaxar is the PAR-1 antagonist drawn here.

BOARD TRAP

Vorapaxar (PAR-1 blocker) is contraindicated with any history of stroke/TIA or ICH because of intracranial bleeding risk — a frequent board contraindication trap.

7. GPIIb/IIIa opens (inside-out signaling)

WHAT YOU SEE

Platelet with GPIIb/IIIa receptors flipping from inactive to active via 'inside-out signaling'.

WHAT IT ENCODES

STEP 4: all agonist pathways converge to activate GPIIb/IIIa (integrin alphaIIbbeta3) through inside-out signaling — a conformational change exposing the fibrinogen binding site. This is the FINAL COMMON PATHWAY of aggregation.

BOARD TRAP

GPIIb/IIIa is the common final pathway — abciximab/eptifibatide/tirofiban block it. Board trap: confusing 'inside-out' activation with adhesion; this step is aggregation (platelet-platelet) not wall adhesion.

8. GPIIb/IIIa cross-linking by fibrinogen

WHAT YOU SEE

Multiple platelets bridged together by VWF/fibrinogen via active GPIIb/IIIa, labeled VWF.

WHAT IT ENCODES

Active GPIIb/IIIa binds fibrinogen (and VWF) to cross-link adjacent platelets — the actual aggregation event building the platelet plug. Final common pathway: every agonist ends here.

BOARD TRAP

Fibrinogen (not fibrin) is the bridging ligand for GPIIb/IIIa aggregation. The fibrin mesh comes later (step 6, thrombin-driven). Don't swap fibrinogen-bridging with fibrin-stabilization.

9. STEP 6 Stabilization — fibrin thrombus

WHAT YOU SEE

A woven basket of fibrin strands locking platelets into a stable white thrombus.

WHAT IT ENCODES

STEP 6 STABILIZATION: thrombin converts fibrinogen to fibrin strands that weave through the aggregate, yielding a stable white (platelet-rich) thrombus. This is the FINAL COMMON PATHWAY shown twice.

BOARD TRAP

White (platelet-rich, arterial) thrombus is treated with antiplatelets; red (fibrin/RBC, venous/stasis) thrombus is treated with anticoagulants. Her annotation contrasts 'white artery / antiPLT' vs 'red vein / fibrin + RBC / anticoag' — a key board discriminator.

10. Prostacyclin (PGI₂) -> cAMP

WHAT YOU SEE

A banner reading PROSTACYCLIN -> increased cAMP at the healthy endothelium baseline.

WHAT IT ENCODES

Healthy endothelium releases prostacyclin (PGI₂), raising platelet cAMP to inhibit activation — a natural antiaggregatory brake. High-dose aspirin can blunt endothelial PGI₂ (the pro-aggregatory shift), which is why low dose is preferred.

BOARD TRAP

Low-dose aspirin spares endothelial PGI₂ (favoring antiplatelet effect); high-dose (~1g/day) also blocks endothelial COX-derived PGI₂ = pro-aggregatory shift. 'More aspirin is better' is the trap.

11. Nitric oxide (NO) -> cGMP

WHAT YOU SEE

A speech bubble labeled NO: nitric oxide -> increased cGMP at the endothelial baseline.

WHAT IT ENCODES

Endothelial nitric oxide raises platelet cGMP, inhibiting activation and causing vasodilation — a second natural antiplatelet defense lost after vascular injury (her note: 'all lost after vascular injury').

BOARD TRAP

NO/cGMP and PGI₂/cAMP are the two endothelial brakes — both are LOST at injured endothelium, which is why thrombosis proceeds there. Don't attribute platelet inhibition here to a drug; these are endogenous.

12. CD39 / ADPase

WHAT YOU SEE

A broom sweeping, labeled CD39 and ADPase along the healthy endothelium banner.

WHAT IT ENCODES

Endothelial CD39 (an ADPase) degrades extracellular ADP, removing the P2Y₁₂ agonist and damping platelet recruitment. Another endogenous antiplatelet defense of intact endothelium.

BOARD TRAP

CD39/ADPase clears ADP — it is endogenous, not a drug. Board trap is calling this clopidogrel's mechanism; clopidogrel blocks the P2Y₁₂ RECEPTOR, it does not enzymatically degrade ADP.

13. Irreversible vs reversible (platelet lifespan)

WHAT YOU SEE

Annotation near step 2/3 noting platelet has no nucleus and inhibition lasts the entire platelet lifespan.

WHAT IT ENCODES

Because platelets lack a nucleus, irreversible drugs (aspirin on COX-1; clopidogrel/prasugrel on P2Y₁₂) inhibit for the platelet lifespan (~7-10 days). Recovery requires new platelet production, hence the 5-7 day pre-surgery hold.

BOARD TRAP

Ticagrelor is REVERSIBLE — its effect wanes with drug half-life, but it is still held ~5 days pre-op. The trap is assuming all P2Y₁₂ inhibitors are irreversible or have identical washout.
